

| Question |     |       |    | Marking details                                                       | Marks Available |          |          |           |          |          |
|----------|-----|-------|----|-----------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |    |                                                                       | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 5        | (a) | (i)   |    | The same as (1)                                                       | 1               |          |          | 1         |          | 1        |
|          |     | (ii)  |    | The same as (1)                                                       | 1               |          |          | 1         |          | 1        |
|          | (b) | (i)   | I  | Current = $4.5/2.25(\text{subs})(1) = 2$ (1)                          | 1               | 1        |          | 2         | 2        | 2        |
|          |     |       | II | Power = $2 (\text{ecf}) \times 4.5 (\text{subs}) (1) = 9\text{W}$ (1) | 1               | 1        |          | 2         | 2        | 2        |
|          |     | (ii)  |    | 4.5V                                                                  |                 | 1        |          | 1         |          | 1        |
|          | (c) | (i)   |    | $A_2$                                                                 |                 | 1        |          | 1         |          | 1        |
|          |     | (ii)  |    | 9V                                                                    |                 | 1        |          | 1         | 1        |          |
|          |     | (iii) | I  | 4A                                                                    |                 | 1        |          | 1         |          |          |
|          |     |       | II | 8A (ecf)                                                              |                 | 1        |          | 1         |          |          |
|          |     |       |    | <b>Question 5 total</b>                                               | <b>4</b>        | <b>7</b> | <b>0</b> | <b>11</b> | <b>4</b> | <b>8</b> |